

Numerical solution of IVP & BVP

1. Picard's method

$$\frac{dy}{dx} = f(x, y)$$

$$\Rightarrow \int_{y_0}^y dy = \int_{x_0}^x f(x, y) \cdot dx$$

$$y - y_0 = \int_{x_0}^x f(x, y) \cdot dx$$

$$y = y_0 + \int_{x_0}^x f(x, y) \cdot dx$$

$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) \cdot dx$$

$$y_2 = y_0 + \int_{x_0}^x f(x, y_1) \cdot dx$$

⋮

$$y_n = y_0 + \int_{x_0}^x f(x, y_n) \cdot dx$$

Problems:

1. Solve the differential equation $\frac{dy}{dx} = x^2 - y$, with $y(0) = 1$
 using Picard's method also find y at $x = 1$

$$f(x, y) = x^2 - y \mid x_0 = 0, y_0 = 1$$

$$\therefore y = y_0 + \int_{x_0}^x f(x, y) \cdot dx$$

$$y = 1 + \int_0^x (x^2 - y) \cdot dx \rightarrow \textcircled{x}$$

$$y_1 = 1 + \int_0^x (x^2 - 1) \cdot dx$$

$$= 1 + \int_0^x (x^2 - 1) \cdot dx$$

$$= 1 + \left[\frac{x^3}{3} - x \right]_0^x \Rightarrow 1 + \frac{x^3}{3} - x$$

$$y_2 = 1 + \int_0^x \left[x^2 - \left(1 + \frac{x^3}{3} - x \right) \right] \cdot dx$$

$$= 1 + \int_0^x \left(x^2 - 1 - \frac{x^3}{3} + x \right) \cdot dx$$

$$= 1 + \frac{x^3}{3} - x - \frac{x^4}{12} + \frac{x^2}{2}$$

$$y_2 = 1 - x + \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{12}$$

$$y_3 = 1 + \int_0^x \left[x^2 - \left(1 - x + \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{12} \right) \right] \cdot dx$$

$$= 1 + \int_0^x \left[x^2 - 1 + x - \frac{x^2}{2} - \frac{x^3}{3} + \frac{x^4}{12} \right] \cdot dx$$

$$= 1 + \frac{x^3}{3} - x + \frac{x^2}{2} - \frac{x^3}{6} - \frac{x^4}{12} + \frac{x^5}{60}$$

$$y_3 = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} - \frac{x^4}{12} + \frac{x^5}{60}$$

Now $x=1$

$$y_3 = 0.6$$

$$② \frac{dy}{dx} = \frac{x^2}{y^2+1}, y(0)=0, \text{ Find } y(1)$$

$$x_0 = 0, y_0 = 0$$

$$y = y_0 + \int_{x_0}^x f(x, y) \cdot dx$$

$$y = 0 + \int_0^x \frac{x^2}{y^2+1} \cdot dx$$

$$y = 0 + \int_0^x \frac{x^2}{y^2+1} \cdot dx \rightarrow (*)$$

$$y = y_0 \text{ for } y_1 \Rightarrow y_1 = \int_0^x x^2 \cdot dx = \frac{x^3}{3}$$

$$\text{Now, } y_2 = \int_0^x \frac{x^2}{\left(\frac{x^3}{3}\right)^2+1} \cdot dx$$

$$\text{Put } \frac{x^3}{3} = t \Rightarrow dt = \frac{3x^2}{3} \cdot dx$$

$$dt = x^2 \cdot dx$$

$$\text{when } x=0, t=0$$

$$x=x, t = \frac{x^3}{3}$$

$$y_2 = \int_0^{\frac{x^3}{3}} \frac{dt}{t^2+1} = \tan^{-1}(t) \Big|_0^{\frac{x^3}{3}}$$

$$y_2 = \tan^{-1}(x^3/3) \Rightarrow \text{at } x=1$$

$$y_2 = \tan^{-1}(1/3)$$

$$y_2 = 0.3217$$

$$\boxed{y(1) = 0.3217}$$

③ $\frac{dy}{dx} = x + y + xy$, $y(0) = 1$ Find $y(0.1)$ go up to

3 situations

$$\Rightarrow x_0 = 0, y_0 = 1$$

$$y = y_0 + \int_{x_0}^x f(x, y) \cdot dx$$

$$= 1 + \int_0^x (x + y + xy) \cdot dx$$

$$y_1 = 1 + \int_0^x (x + 1 + x) \cdot dx \quad \text{here } y \text{ becomes } y_0$$

$$y_1 = 1 + \left[\frac{x^2}{2} + x + \frac{x^2}{2} \right]_0^x$$

$$= 1 + \left[x^2 + x \right]_0^x$$

$$\boxed{y_1 = 1 + x^2 + x}$$

$$y_2 = 1 + \int_0^x (x + y_1 + xy_1) \cdot dx$$

$$= 1 + \int_0^x (x + 1 + x^2 + x + x(1 + x^2 + x)) \cdot dx$$

$$= 1 + \int_0^x (x + 1 + x^2 + x + x + x^3 + x^2) \cdot dx$$

$$= 1 + \int_0^x (1 + 3x + 2x^2 + x^3) \cdot dx$$

$$= 1 + \left[x + \frac{3x^2}{2} + \frac{2x^3}{3} + \frac{x^4}{4} \right]_0^x$$

$$y_2 = 1 + \left[x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} \right]$$

$$y_3 = 1 + \int_0^x (x + y_2 + x y_2)$$

$$= 1 + \int_0^x \left(x + 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} + x \left(1 + x + \frac{3}{2}x^2 + \right. \right.$$

$$\left. \frac{2}{3}x^3 + \frac{x^4}{4} \right)$$

$$= 1 + \int_0^x \left(x + 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} + x + x^2 + \frac{3}{2}x^3 + \frac{2}{3}x^4 + \frac{x^5}{5} \right)$$

$$= 1 + \left[\frac{x^2}{2} + x + \frac{x^2}{2} + \frac{3}{2} \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{20} + \frac{x^2}{2} + \frac{x^3}{3} + \frac{3}{2} \frac{x^4}{4} + \frac{2}{3} x \frac{x^5}{5} + \frac{x^6}{30} \right]_0^x$$

$$= 1 + \frac{x^2}{2} + x + \frac{x^2}{2} + \frac{3x^3}{2} + \frac{x^4}{6} + \frac{x^5}{20} + \frac{x^2}{2} + \frac{x^3}{3} + \frac{3x^4}{8} + \frac{2x^5}{15} + \frac{x^6}{30}$$

$$y_3 = 1 + x + \frac{3}{2}x^2 + \frac{5}{6}x^3 + \frac{13}{24}x^4 + \frac{11}{60}x^5 + \frac{x^6}{30}$$

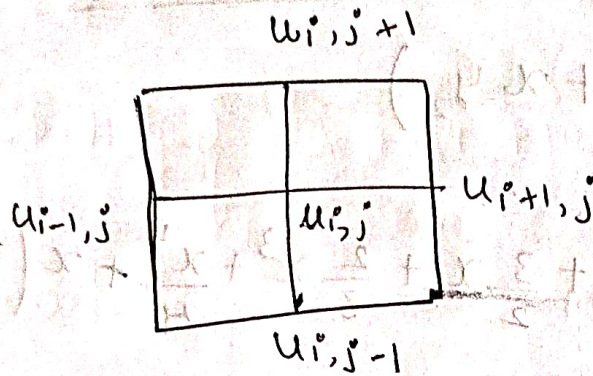
$$y_3 = 1.01159$$

④ $\frac{dy}{dx} = \frac{y-x}{y+x}$ $y(0) = 1$, $y(0.1)$

09/04/25

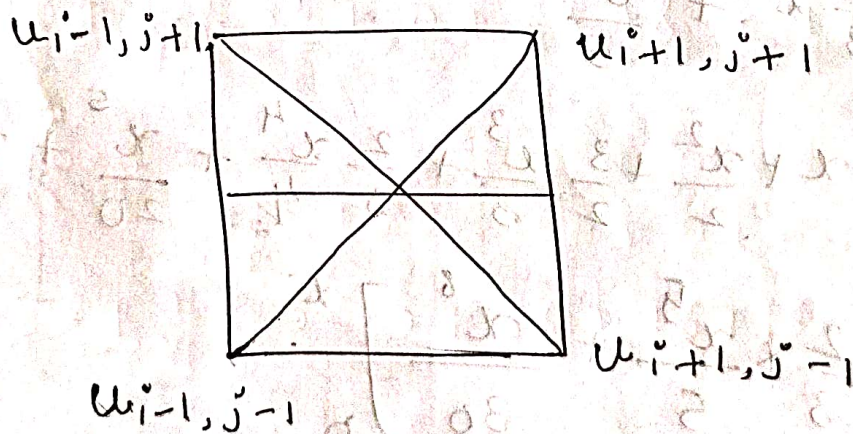
$x_0 = 0$, $y_0 = 1$

Successive over relaxation method :-



$$u_{ij}^o = \frac{1}{4} (u_{i,j-1} + u_{i,j+1} + u_{i-1,j} + u_{i+1,j})$$

standard five point formula



$$u_{ij}^o = \frac{1}{4} \{ u_{i-1,j+1} + u_{i+1,j+1} + u_{i-1,j-1} + u_{i+1,j-1} \}$$

Diagonal 5-Point formula

$$u_{ij}^{n+1} = u_{ij}^{(n)} + \frac{w}{4} \{ u_{i-1,j+1}^{(n+1)} + u_{i+1,j+1}^{(n)} + u_{i-1,j-1}^{(n)} + u_{i+1,j-1}^{(n)} - 4u_{ij}^{(n)} \}$$

If w not given $w = 1.0$ } $\rightarrow$ default

1. Solve the equation $u_{xx} + u_{yy} = 0$ by SOR method inside the square bounded by the lines $x=0, y=0, x=4, y=4$ and $u = x^2 y^2$ on the boundary $\{h=1, k=1\}$ to y

↑ $y=4$ 0 16 64 144 256

$y=3$ 0 24 u_1 56 u_2 u_3 104

$y=2$ 0 16 u_4 u_5 32 u_6 56

$y=1$ 0 8 u_7 u_8 16 u_9 24

$y=0$ 0 0 0 0 0

$x=0$ $x=1$ $x=2$ $x=3$ $x=4$

corresponds to x

take $w=1$

to find boundary $u = x^2 y^2$

$$u_5 = \frac{1}{4} (0 + 64 + 64 + 0) = 32$$

$$u_1 = \frac{1}{4} (0 + 0 + 64 + 32) = 24$$

$$u_3 = \frac{1}{4} (64 + 256 + 32 + 64) = 104$$

$$u_7 = \frac{1}{4} (0 + 0 + 32 + 0) = 8$$

$$u_9 = \frac{1}{4} (32 + 64 + 0 + 0) = 24$$

$$u_2 = \frac{64 + 32 + 24 + 104}{4} = 56$$

$$u_4 = \frac{1}{4} (24 + 32 + 8 + 32) = 16$$

$$u_6 = \frac{1}{4} (104 + 32 + 64 + 24) = 56$$

$$u_8 = \frac{1}{4} (32 + 8 + 24 + 0) = 16$$

$$u_{i,j}^{(n+1)} = u_{i,j}^{(n)} + \frac{w}{4} \{ \text{four points} - 4(u_{i,j})^n \}$$

$$u_1^{(1)} = 24 + \frac{1}{4} \{ 0 + 16 + 56 + 16 - 4 \times 24 \}$$

$$= 22$$

$$u_2^{(1)} = 56 + \frac{1}{4} \{ 22 + 64 + 104 + 32 - 4 \times 56 \}$$

$$u_2^{(1)} = 55.5$$

$$u_3^{(1)} = 104 + \frac{1}{4} \{ 55.5 + 244 + 144 + 56 - 4 \times 104 \}$$

$$u_3^{(1)} = 99.8750$$

$$u_4^{(1)} = 16 + \frac{1}{4} \{ 22 + 0 + 32 + 8 - 4 \times 16 \}$$

$$u_4^{(1)} = 15.5$$

$$u_5^{(1)} = 32 + \frac{1}{4} \{ 55.5 + 56 + 16 + 15.5 - 4 \times 32 \}$$

$$u_5^{(1)} = 35.75$$

$$u_6^{(1)} = 56 + \frac{1}{4} \{ 99.875 + 64 + 35.75 + 24 - 4 \times 56 \}$$

$$u_6^{(1)} = 55.9063$$

$$u_7^{(1)} = 8 + \frac{1}{4} \left\{ 0 + 0 + 16 + 15.5 - 4 \times 8 \right\} =$$

$$u_7^{(1)} = 7.8750$$

$$u_8^{(1)} = 16 + \frac{1}{4} \left\{ 7.8750 + 0 + 35.75 + 24 - 4 \times 16 \right\}$$

$$u_8^{(1)} = 16.9063$$

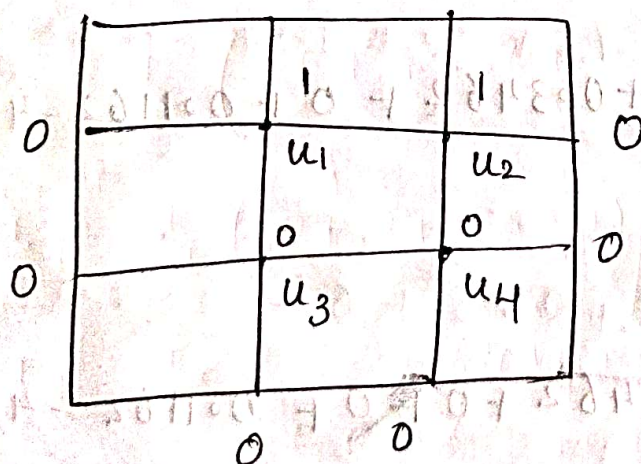
$$u_9^{(1)} = 24 + \frac{1}{4} \left\{ 16 + 0 + 16.9063 + 55.9063 - 4 \times 24 \right\}$$

$$u_9^{(1)} = 22.2032$$

15/04/25

2. Solve the equation $u_{xx} + u_{yy} = 0$ in the domain of the figure below by SOR method by considering $\omega = 1.1$ carryout two iterations

Soln:



write the maximum value for

u_1, u_2, u_3, u_4

1st iteration (SOR)

$$u_{ij}^{(n+1)} = u_{ij}^{(n)} + \frac{\omega}{H} \left[u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)} + u_{i+1,j}^{(n)} + u_{i-1,j}^{(n)} - 4 \cdot u_{ij}^{(n)} \right]$$

$$u_1^{(1)} = 1 + \frac{1 \cdot 1}{4} [0 + 1 + 1 + 0 - 4(1)] = \boxed{u_1^{(1)} = 0.45}$$

$$u_2^{(1)} = 1 + \frac{1 \cdot 1}{4} [0.45 + 1 + 0 + 0 - 4(1)] = \boxed{u_2^{(1)} = 0.2988}$$

$$u_3^{(1)} = 0 + \frac{1 \cdot 1}{4} [0.45 + 0 + 0 + 0] = \boxed{u_3^{(1)} = 0.1238}$$

$$u_4^{(1)} = 0 + \frac{1 \cdot 1}{4} [0.2988 + 0.1238 + 0 - 0]$$

$$\boxed{u_4^{(1)} = 0.1162}$$

2nd iteration

$$u_1^{(2)} = 0.45 + \frac{1 \cdot 1}{4} [1 + 0 + 0.2988 + 0.1238 - 4(0.45)]$$

$$\boxed{u_1^{(2)} = 0.3462}$$

$$u_2^{(2)} = 0.2988 + \frac{1 \cdot 1}{4} [1 + 0.3462 + 0 + 0.1162 - 4 \times 0.2988]$$

$$\boxed{u_2^{(2)} = 0.3723}$$

$$u_3^{(2)} = 0.1238 + \frac{1 \cdot 1}{4} [0.3462 + 0 + 0 + 0.1162 - 4 \times 0.1238]$$

$$\boxed{u_3^{(2)} = 0.1148}$$

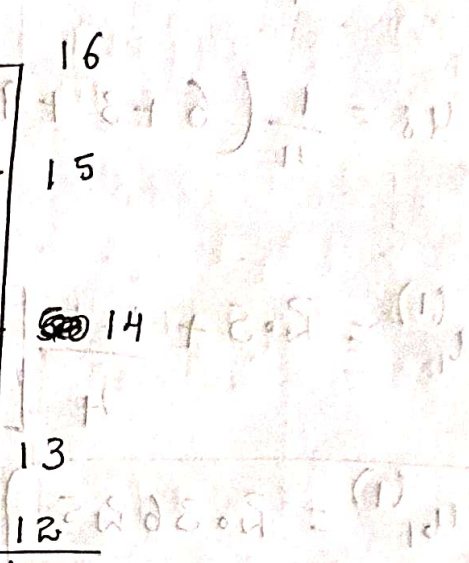
$$u_4^{(2)} = 0.1162 + \frac{1 \cdot 1}{4} [0.3723 + 0.1148 + 0 + 0 - 4 \times 0.1162]$$

$$\boxed{u_4^{(2)} = 0.1223}$$

$$g(y) = 12 + y \quad 0 \leq y \leq 4$$

$\frac{1}{1} = 1$
 $\frac{1}{1} = 1$

11



$$\frac{101422}{13} = 78016.46$$

6 $\Rightarrow$ $u_5 = 6$

$$F = \frac{1}{11} \cdot \frac{1}{10} \cdot \frac{1}{10} = \frac{1}{1100}$$

13. 10. 1913

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$$u_4 = \frac{1}{4} (0 + 6 + 3 + 2.5) = 2.8750$$

$$u_6 = \frac{1}{4} (10 + 6 + 14 + 9.5) = 9.8750$$

$$u_8 = \frac{1}{4} (6 + 3 + 9.5 + 6) = 6.1250$$

$$u_1^{(1)} = 2.5 + \frac{1.1}{4} [1 + 0 + 5.625 + 2.875] - 4 \times 2.5$$

$$u_1^{(1)} = 2.3625$$

$$u_2^{(1)} = 5.625 + \frac{1.1}{4} [4 + 2.3625 + 10 + 6] - 4 \times 5.625$$

$$u_2^{(1)} = 5.5872$$

$$u_3^{(1)} = 10 + \frac{1.1}{4} [9 + 5.5872 + 15 + 9.875] - 4 \times 10$$

$$u_3^{(1)} = 9.9313$$

$$u_4^{(1)} = 2.875 + \frac{1.1}{4} [2.3625 + 0 + 3 + 6] - 4 \times 2.875$$

$$u_4^{(1)} = 2.8372$$

$$u_5^{(1)} = 6 + \frac{1.1}{4} [5.5872 + 2.8372 + 9.875 + 6.125] - 4 \times 6$$

$$u_5^{(1)} = 6.1167$$

$$u_6^{(1)} = 9.875 + \frac{1.1}{4} [6.1167 + 9.9313 + 14 + 9.5] - 4 \times 9.875$$

$$u_6^{(1)} = 9.8882$$

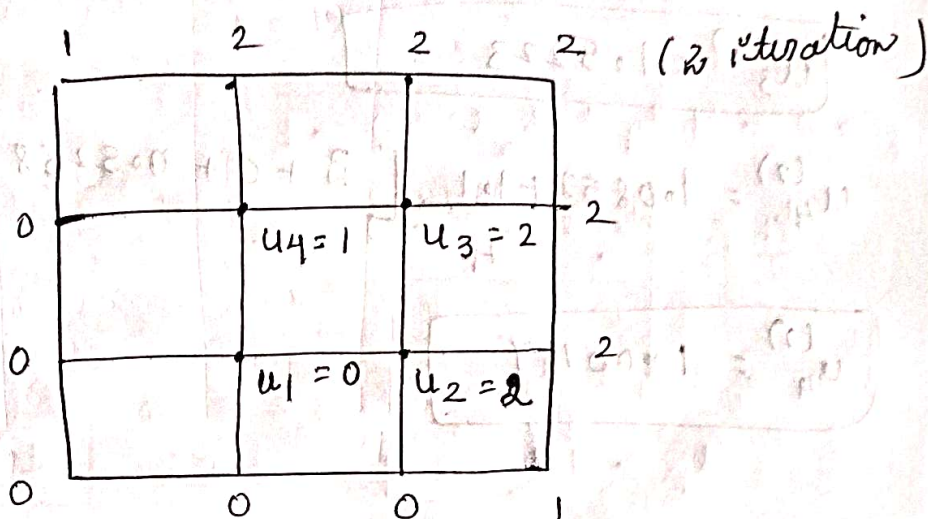
$$u_7^{(1)} = 3 + \frac{1.1}{4} [2.8372 + 3 + 0 + 6.1250] = \boxed{2.9896 = u_7^{(1)}}$$

$$u_8^{(1)} = 6.1250 + \frac{1.1}{4} [6.1167 + 6 + 2.9896 + 9.5] = \boxed{6.7942 = u_8^{(1)}}$$

$$u_9^{(1)} = 9.5 + \frac{1.1}{4} [9.8882 + 9 + 13 + 6.1542 - 4 \times 9.5]$$

$$\boxed{u_9^{(1)} = 9.5117}$$

4) solve $\nabla^2 u = 0$, for



1st iteration

$$u_1^{(1)} = 0 + \frac{1.1}{4} [0 + 0 + 2 + 1] = u_1^{(1)} = 0.8250 = \boxed{u_1^{(1)} = 0.8250}$$

$$u_2^{(1)} = 2 + \frac{1.1}{4} [0.8250 + 0 + 2 + 2] \Rightarrow \boxed{u_2^{(1)} = 3.3269}$$

$$u_3^{(1)} = 2 + \frac{1.1}{4} [1.1269 + 1 + 2 + 2 - 4 \times 2] \Rightarrow \boxed{u_3^{(1)} = 1.4849}$$

$$u_4^{(1)} = 1 + \frac{1.1}{4} [2 + 0 + 0.8250 + 1.4849 - 4 \times 1]$$

$$\boxed{u_4^{(1)} = 1.0852}$$

2nd iteration

$$u_1^{(2)} = 0.8250 + \frac{1.1}{4} [0 + 0 + 1.1269 + 1.0852]$$

$$- 4 \times 0.8250$$

$$\boxed{u_1^{(2)} = 0.5258}$$

$$u_2^{(2)} = 1.01269 + \frac{1.01}{4} [0.5258 + 0 + 2 + 1.4849 - 4 \times 1.01269]$$

$$u_2^{(2)} = 0.9903$$

$$u_3^{(2)} = 1.04849 + \frac{1.01}{4} [2 + 2 + 0.9903 + 1.0852 - 4 \times 1.04849]$$

$$u_3^{(2)} = 1.05223$$

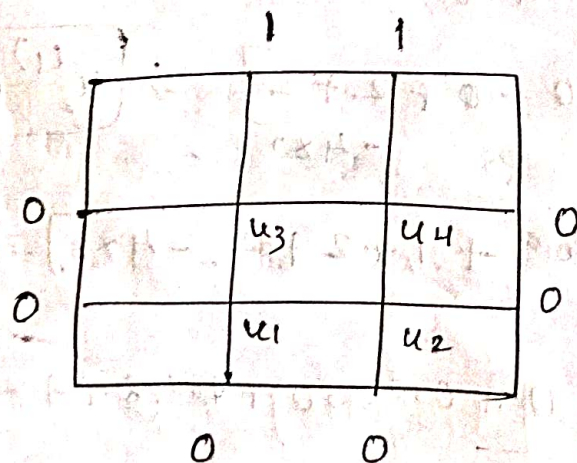
$$u_4^{(2)} = 1.0852 + \frac{1.01}{4} [2 + 0 + 0.5258 + 1.05223 - 4 \times 1.0852]$$

$$u_4^{(2)} = 1.00047$$

16/04/25

ADI method [Alternating direction implicit method]

to solve $\nabla^2 u = 0$



$$u_{i-1,j}^{(\tau+1)} - 4u_{i,j}^{(\tau+1)} + u_{i+1,j}^{(\tau+1)} = -u_{i,j-1}^{(\tau)} - u_{i,j+1}^{(\tau)} \rightarrow \textcircled{1}$$

$$u_{i,j+1}^{(\tau+2)} - 4u_{i,j}^{(\tau+2)} + u_{i,j-1}^{(\tau+2)} = -u_{i-1,j}^{(\tau+1)} - u_{i+1,j}^{(\tau+1)} \rightarrow \textcircled{2}$$

$$0 - 4u_1 + u_2 = -0 - 1$$

$$u_1 - 4u_2 + 0 = -0 - 1$$

$$\Rightarrow -4u_1 + u_2 = -1$$

$$u_1 - 4u_2 = -1$$

$$u_1^{(1)} = u_2^{(1)} = 1/3$$

	1	1
0		
	1/3	1/3
0		
	1/3	1/3

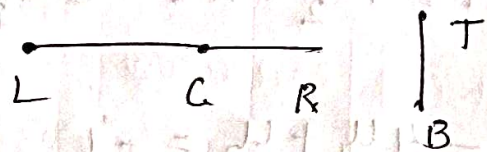
$$0 - 4u_3 + u_4 = -0 - 1$$

$$u_3 - 4u_4 + 0 = -0 - 1$$

$$\Rightarrow -4u_3 + u_4 = -1$$

$$u_3 - 4u_4 = -1$$

$$u_3^{(1)} = 1/3 = u_4^{(1)}$$



$$0 - 4u_1 + u_3 = 0 - 1/3$$

$$u_1 - 4u_3 + 1 = 0 - 1/3$$

$$\Rightarrow -4u_1 + u_3 = -1/3$$

$$u_1 - 4u_3 = -1/3$$

$$u_1^{(2)} = 0.1778$$

$$u_3^{(2)} = 0.3778$$

	1	1
0		
	0.3778	0.3778
0		
	0.1778	0.1778

$$0 - 4u_2 + u_4 = -1/3 - 0$$

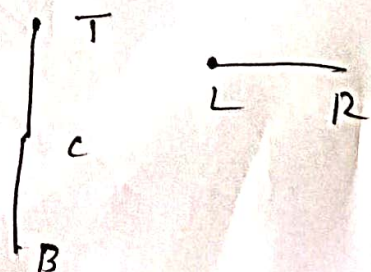
$$u_2 - 4u_4 + 1 = -1/3 - 0$$

$$\Rightarrow -4u_2 + u_4 = -1/3$$

$$u_2 - 4u_4 = -1/3$$

$$u_2^{(2)} = 0.1778$$

$$u_4^{(2)} = 0.3778$$



2. Solve $u_{xx} + u_{yy} = 0$, given

	2	2
0		
0	$u_3 = 1$	$u_4 = 2$
	$u_1 = 0$	$u_2 = 1$

1st iteration

$$0 - 4u_1 + u_2 = -0 - 1$$

$$u_1 - 4u_2 + 2 = -0 - 2$$

$$\Rightarrow -4u_1 + u_2 = -1$$

$$u_1 - 4u_2 = -4$$

$$u_1^{(1)} = 0.5333$$

$$u_2^{(1)} = 1.1333$$

$$0 - 4u_3 + u_4 = -0 - 2$$

$$u_3 - 4u_4 + 2 = 1 - 2$$

$$\Rightarrow -4u_3 + u_4 = -2$$

$$u_3 - 4u_4 = -5$$

$$u_3^{(1)} = 0.8667$$

$$u_4^{(1)} = 1.4667$$

	2	2
0		
0	$u_3 = 0.8667$	$u_4 = 1.4667$
	$u_1 = 0.5333$	$u_2 = 1.1333$

2nd iteration

$$0 - 4u_1 + u_3 = -0 - 1.1333$$

$$u_1 - 4u_3 + 2 = -0 - 1.4667$$

$$\Rightarrow -4u_1 + u_3 = -1.1333$$

$$u_1 - 4u_3 = -3.4667$$

$$u_1^{(2)} = 0.5333$$

$$u_3^{(2)} = 1.0000$$

$$0 - 4u_2 + u_4 = -0.5333 - 2$$

$$u_2 - 4u_4 + 2 = -0.8667 - 2$$

$$\Rightarrow -4u_2 + u_4 = -2.5333$$

$$u_2 - 4u_4 = -4.8667$$

$$u_2^{(2)} = 1$$

$$u_4^{(2)} = 1.4667$$

	1	2	2
0		$u_3 = 1$	$u_4 = 1.4667$
0		$u_1 = 0.5333$	$u_2 = 1$
	0	0	0